



FinaBridge

Option D — Facilitator Model

AI Document Verification + 3-Tier Admin Review Pipeline

CHAIRMAN BRIEFING DOCUMENT

STRICTLY CONFIDENTIAL

PREPARED FOR

DIVISION

REFERENCE

VERSION

DATE

CLASSIFICATION

2 min	3 Tiers	13	100%	5 Phases
AI Verification	Human Review	Email Touchpoints	Audit Coverage	Trade Lifecycle

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FinaBridge Option D — Chairman Briefing | Reference: FTD-BRIDGE-OPT-D-2025

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01 Executive Summary

Strategic overview — FinaBridge Option D Facilitator Model

FinaBridge Option D transforms Finatrades from a passive marketplace into an active trade finance facilitator. Every trade request undergoes AI-powered document verification and structured three-tier human review before being published — ensuring institutional-grade compliance, fraud prevention, and risk management on every transaction.

Ø>Y

AI Document Verification

OpenAI GPT-4o Vision reads LC, POL and WR documents within minutes of submission. Extracts all critical fields and generates a fraud risk score from 0–100.

Ø<RUB

Three-Tier Human Review

Every AI-passed document undergoes review by Macy (Operations), Farah (Compliance), and Director Reda (Final Authority) before any trade is published to exporters.

Ø=Y

Gold-Locked Settlement

Upon proposal acceptance, importer gold is locked in a cryptographic Settlement Hold. Released only after goods delivery confirmation and admin authorisation.

Ø=U

Secure Deal Room

Encrypted real-time deal room with role-based access for importer, exporter and admin. Complete document exchange, messaging and audit trail permanently archived.

~2 min	3 Layers	13 Emails	USD 50K+
AI Verification Speed	Human Review Depth	Notification Touchpoints	Per-Trade Capacity

Business Objective

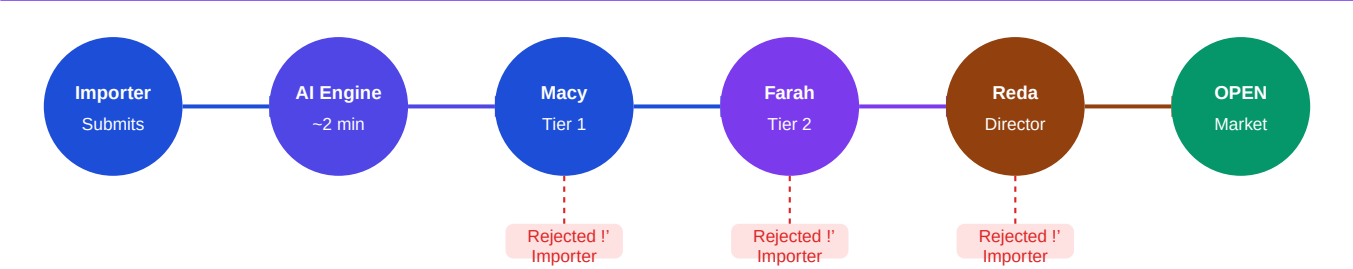
Position Finatrades as a trusted facilitator in international trade finance, generating fee revenue on every transaction while maintaining institutional-grade compliance and risk management standards.

02 Review Team & Responsibilities

Three-tier human review pipeline — accountability and escalation structure

Macy Operations Analyst ● First-line document review ● Verify AI-extracted data against original document ● Confirm bank stamp and signature authenticity ● Cross-check amounts, dates and beneficiary name ● Approve !' Tier 2 or Reject with written reason	Farah Compliance Officer ● Full AML (Anti-Money Laundering) screening ● Sanctions list and PEP check on both parties ● Company registration and licence verification ● KYC cross-reference with submitted documents ● Approve !' Director or Reject with compliance reason	Reda Director, Trade Finance ● Reviews complete audit trail (AI + Macy + Farah) ● Makes final business and risk judgement ● Approves !' Publishes trade to live marketplace ● Rejects with Director-level reason for importer ● Full override authority at any review stage
OWNS: Operations integrity — ensures AI output is accurate before compliance review	OWNS: Regulatory compliance — ensures every trade meets legal and AML/KYC standards	OWNS: Final authority — business decision, risk appetite and marketplace publication

Escalation Flow & Decision Points



03 Phase 1 — Trade Submission

Importer creates trade request with payment instrument selection and document upload

The Story — Ahmed Al-Rashid (Importer, Dubai)

Ahmed is a Dubai-based electronics importer. He logs into Finatrades and navigates to FinaBridge. As a first-time user, a Terms & Conditions Disclaimer Modal appears. He selects "Importer" as his role, digitally signs with his full name, and clicks Accept. He is now inside the FinaBridge dashboard.

He clicks "Create Trade" and works through the structured submission form in five sections:

Section A — Payment Instrument

Payment Instrument Type *

Letter of Credit (LC) %Ä NEW FIELD — replaces old "Finance Type"

Supporting Document Upload *

LC_BankOfChina_2025.pdf (10MB max — conditional on LC/POL/WR selection)

Section B — Goods & Trade Value

Goods Name

Electronic Components

Quantity

5,000 Units

Trade Value (USD)

USD 50,000.00

Gold Settlement

15.234 grams (auto-calculated from live gold price)

Currency

USD

Payment Terms

30 days from Bill of Lading date

Section C — Shipping Details

Mode of Transport

Sea

Port of Loading

Shanghai, China

Port of Discharge

Jebel Ali, Dubai, UAE

Incoterms

FOB

Expected Ship Date

15 August 2025

Partial Shipment

Not Allowed

Transshipment

Not Allowed

Section D — Required Documents (from Exporter)

Required Documents

' Bill of Lading ' Commercial Invoice ' Packing List

' Certificate of Origin ' Insurance Certificate

Section E — Exporter Preference

Exporter Known?

No — Platform to suggest matching exporters

Key Change from Current System

"Finance Type" field (previously a dead dropdown with only one option) is replaced with "Payment Instrument Type" — LC / POL / WR / FinaBridge Wallet, each with conditional document upload.

04 Phase 2 — AI Verification Engine

Automated document analysis, field extraction and fraud scoring — ~2 minutes

What Happens After Ahmed Clicks Submit

Immediately upon submission, a background job is queued. The AI verification engine runs completely transparently — Ahmed sees "Under Review" status while the system works:

- 1

Document Retrieved

LC PDF downloaded from Cloudflare R2 secure encrypted storage and prepared for analysis
- 2

Vision Analysis

OpenAI GPT-4o Vision processes the full document — reads all text, stamps, tables, signatures and seals
- 3

Field Extraction

Structured data extracted: Issuing Bank, LC Number, Amount, Currency, Expiry Date, Beneficiary Name, Goods Description, Incoterms
- 4

Fraud Analysis

Six automated checks run in parallel: Amount vs declared value, Expiry date validation, KYC name match, Bank registry verification, Document consistency, Format authenticity
- 5

Score Generated

Fraud Risk Score calculated 0–100. Score below 50 = PASS. Score 50 or above = FAIL with specific flags identified
- 6

Routing Decision

PASS !' Status updates to "Tier 1 Review" !' Emails sent to Macy + Farah + Reda + Ahmed. FAIL !' Status = "AI Rejected" !' Rejection emails sent immediately.

Sample AI Extraction — Ahmed's LC Document

AI VERIFICATION PASSED			100
Issuing Bank	Bank of China, Shanghai Branch	Verified Institution	
LC Number	LC2025-88442	Valid Format	
Amount	USD 50,000	Matches Trade Value	
Expiry Date	30 September 2025	Future Date	
Beneficiary	Ahmed Al-Rashid Trading LLC	KYC Name Match	
Goods	Electronic Components, 5,000 units	Consistent with Submission	

If AI FAILS (alternate path)

Score "e 50 triggers immediate rejection. Importer receives detailed email listing each flag.
Example flags: Amount mismatch detected (LC: \$30,000 vs declared \$50,000) | Document expired (15 Jan 2025)
Macy, Farah and Reda all receive an FYI alert email. No admin action required on AI rejections.

05 Phase 3 — Three-Tier Admin Review

Sequential approval pipeline: Macy ! Farah ! Director Reda

Macy — Tier 1: Operations Review

Macy receives an email with "ACTION REQUIRED" subject line. She opens the Admin Panel ! FinaBridge ! Review Queue ! Tier 1. Ahmed's trade card shows the AI fraud score of 8/100 and all extracted fields. She opens the LC document in a side-by-side viewer with the AI report.

- ' Bank stamp is clear, authentic and matches issuing bank in AI extraction
- ' Amount of USD 50,000 visible on document matches trade declaration
- ' Beneficiary name "Ahmed Al-Rashid Trading LLC" matches KYC on file
- ' LC expiry date (30 Sep 2025) is well within trading window

DECISION: Macy clicks "Approve ! Tier 2" and adds note: "Document verified. Bank stamp clear. Amount and beneficiary confirmed."

EMAIL: Farah receives ACTION REQUIRED email | Reda + Macy receive CC awareness copy

Farah — Tier 2: Compliance Review

Farah receives the Tier 1 approval email with Macy's notes attached. She opens the Tier 2 queue and sees the complete review trail alongside the original document and AI report. She runs formal compliance checks.

- ' AML screening completed — no suspicious transaction patterns identified
- ' Sanctions list check — neither Ahmed nor Wang Electronics appears on any list
- ' Company registration confirmed against UAE trade registry database
- ' KYC documents cross-referenced — all details consistent

DECISION: Farah clicks "Approve ! Director" and notes: "AML clear. No sanctions match. Company registration confirmed against UAE trade registry."

EMAIL: Reda receives FINAL DECISION REQUIRED email | Farah + Macy receive CC awareness copy

Reda — Director: Final Approval

Reda receives the escalation email showing the complete review trail: AI passed (8/100), Macy approved, Farah approved. He opens the Final Approval panel and sees the full picture — document, AI report, both reviewer notes — in a single view.

- ' Full audit trail complete: AI ' Macy ' Farah '
- ' Trade value (USD 50,000) within approved single-transaction limits
- ' Instrument type (LC) from verified issuing bank
- ' No business or risk concerns identified

DECISION: Reda clicks "Approve — Publish to Marketplace". Trade status immediately changes to OPEN. publishedToExporters = true.

EMAIL: Ahmed receives "Trade is LIVE" email | Macy + Farah receive FYI completion email

Rejection at any tier terminates the pipeline

Importer receives a detailed rejection email with the specific reason and guidance for resubmission.

The rejection decision is permanently logged in the audit trail. Reda is informed of all rejections.

06 Phase 4 — Marketplace & Exporter Proposals

Trade published live, exporter discovers and submits a structured proposal

The Story — Trade Goes Live

The moment Reda approves, Ahmed's trade is instantly published to the FinaBridge marketplace. Status changes to "OPEN". Ahmed receives a congratulatory email. The trade card appears to all registered exporters in the Browse Marketplace section.

Wang Electronics Co. Ltd — Exporter, Shanghai

David Wang is a Shanghai-based electronics manufacturer. He logs into Finatrades, switches to "Exporter" mode in FinaBridge, and opens the Marketplace tab. He filters by "Sea Freight" and "FOB" and finds Ahmed's trade:

TR-2025-0847 Electronic Components 5,000 Units		% OPEN
Trade Value	USD 50,000 Gold Settlement: 15.234g	
Route	Shanghai !' Jebel Ali, Dubai FOB Sea Freight	
Docs Required	B/L, Invoice, Packing List, COO, Insurance	

Wang Submits a Proposal

Wang clicks "Submit Proposal" and fills in the structured proposal form:

Quote Price	USD 48,500
Timeline	35 Days from order confirmation
Port of Loading	Shanghai Yangshan Deep-Water Port
Shipping Method	Sea — Full Container Load (FCL)
Incoterms	FOB
Estimated Delivery	28 August 2025
Insurance	Included in quote price
Certifications	ISO 9001:2015, CE Marking, RoHS Compliant
Company Name	Wang Electronics Co. Ltd
Reg. Number	CN-SH-20180045
Contact Person	David Wang, Director of Export Operations
Contact Email	david@wangelectronics.cn
Notes	Exported similar components to GCC 12 times. All certifications immediately available.
Attachment	WangElectronics_CompanyProfile_2025.pdf

07 Phase 5 — Proposal Acceptance & Gold Lock

Importer accepts, gold is cryptographically reserved, Deal Room created

Ahmed Reviews Wang's Proposal

Ahmed receives an email notification that a new proposal has arrived. He logs in and navigates to My Requests !' TR-2025-0847 !' View Proposals. The platform displays admin-forwarded and shortlisted proposals. Ahmed reviews Wang's offer:

Criteria	Wang's Offer	Ahmed's Requirement	Assessment
Quote Price	USD 48,500	USD 50,000 (value)	Wang offers USD 1,500 savings
Timeline	35 days	Standard 30–45 days	Within acceptable range
Certifications	ISO 9001, CE, RoHS	Required	All certifications met
Insurance	Included	Required	Covered, no extra cost
Port	Yangshan, Shanghai	Shanghai area	Correct loading port

Ahmed Accepts — Three Things Happen Simultaneously

- 1

Gold Reserved
15.234 grams of gold immediately locked from Ahmed's FinaBridge Wallet into a cryptographic Settlement Hold. Gold cannot be accessed, withdrawn or transferred until the hold is released or cancelled by admin.
- 2

Deal Room Created
A secure Deal Room is instantiated with three participants: Ahmed (Importer), David Wang (Exporter), and an assigned Finatrades Admin. Real-time WebSocket connection established.
- 3

Status Updated
Trade request status changes from "Proposal Review" to "Active Trade". All other pending proposals for this trade are automatically declined. Wang's proposal status changes to "Accepted".

Settlement Hold — Gold Custody
The 15.234g gold is not held by either party during the active trade period.
It is locked in Finatrades's settlement infrastructure — neutral custody — until the admin releases it.
This protects both parties: exporter knows payment is guaranteed; importer knows gold cannot leave until delivery confirmed.

08 Phase 6 — Deal Room Operations

Secure real-time communication, document exchange and admin oversight

Entering the Deal Room

Both Ahmed and David Wang enter the Deal Room for the first time and are presented with the Deal Room Terms & Conditions modal covering: Confidentiality, AML/KYC compliance, DIAC arbitration, gold-backed settlement acknowledgement, document authenticity warranty, and UAE governing law. Both must tick "I agree" and click Accept before accessing the chat.

Typical Deal Room Conversation (4–6 Weeks)

David Wang (Exporter) Shipment booking confirmed. Vessel: EVER GIVEN II. ETD Shanghai: August 5, 2025. Uploading Draft B/L for your review.	Aug 1
David Wang (Exporter) 📎 [DOCUMENT ATTACHED: Draft_BL_TR2025-0847.pdf]	Aug 1
Ahmed Al-Rashid (Importer) Draft B/L received. Reviewing... One correction needed: consignee name should read "Ahmed Al-Rashid Trading LLC" — currently shows "Ahmed Trading LLC". Please correct and reissue.	Aug 2
David Wang (Exporter) Correction noted. Revised B/L issued by carrier. Uploading now.	Aug 3
David Wang (Exporter) 📎 [DOCUMENT ATTACHED: Revised_BL_FINAL_TR2025-0847.pdf]	Aug 3
Ahmed Al-Rashid (Importer) Revised B/L confirmed '. Consignee name correct. Approved.	Aug 4
David Wang (Exporter) 📎 [DOCUMENT ATTACHED: Commercial_Invoice_\$48500.pdf] [DOCUMENT ATTACHED: PackingList_5000units.pdf]	Aug 4
Finatrades Admin Official Notice: All required documents received. Commercial Invoice, Packing List and B/L verified. Awaiting Certificate of Origin and Insurance Certificate for full document set completion.	Aug 5
David Wang (Exporter) 📎 [DOCUMENT ATTACHED: CertificateOfOrigin_China.pdf] [DOCUMENT ATTACHED: MarineInsurance_Certificate.pdf]	Aug 6
Finatrades Admin Official Notice: Complete document set received and verified. All 5 required documents confirmed authentic. Awaiting vessel arrival and importer receipt confirmation.	Aug 7
Ahmed Al-Rashid (Importer) Vessel arrived Jebel Ali Port. Goods inspected — 5,000 units of Electronic Components received. All items match packing list. Quality verified. Delivery confirmed '	Sep 12

09 Phase 7 — Gold Settlement & Deal Room Closure

Admin processes gold release, trade completed, archive preserved permanently

Settlement Processing — Admin Action

Following Ahmed's delivery confirmation in the Deal Room, the Finatrades Admin navigates to FinaBridge Management !' Settlement Holds and locates TR-2025-0847. The admin verifies all conditions are met:

- All 5 required documents uploaded, reviewed and verified in Deal Room
- Importer has confirmed receipt of goods in the Deal Room chat (12 Sep 2025)
- No disputes raised by either party within the 48-hour review window
- Settlement Hold status: Active — 15.234g gold reserved and available for release

- 1

Admin Clicks "Release Settlement"
Single action triggers the full settlement cascade. No manual gold transfer required.
- 2

Gold Debited from Settlement Hold
15.234g automatically removed from the cryptographic Settlement Hold (neutral custody).
- 3

Platform Fee Deducted
e.g., 1% platform fee = 0.152g deducted from settlement before transfer to exporter.
- 4

Gold Credited to Exporter
15.082g (net of fee) credited to David Wang's FinaBridge Wallet. Immediately available.
- 5

Transaction Records Created
Full transaction entries created for both Ahmed and Wang. Visible in their transaction histories.
- 6

Trade Status !' Completed
Trade request status updated to "Completed". Timestamp recorded. Audit trail finalised.

Deal Room Closure — Final Step

After settlement is confirmed, the admin formally closes the Deal Room:

- Admin clicks "Close Deal Room" from inside the Deal Room interface
- Closure notes added: "Trade settled. All documents verified. Gold transferred 15 Sep 2025."
- Deal Room immediately becomes read-only — no new messages can be sent by any party
- Both Ahmed and David Wang receive "Deal Room Closed" email with archive link
- Full message history and all documents remain accessible for 7 years (regulatory requirement)

Trade TR-2025-0847 — Complete

Submitted: 22 March 2025 | Approved: 22 March 2025 | Matched: 25 March 2025 | Active: 26 March 2025 | Completed: 15 September 2025
Total Trade Duration: ~176 days | Gold Settled: 15.082g net | Platform Fee Collected: 0.152g

10 Complete Email Notification Map

All 13 automated email triggers across the full trade lifecycle

Every email is triggered automatically by a status change or user action. No manual sending required. Admin team (Macy, Farah, Reda) and both parties (Importer, Exporter) receive timely, targeted notifications at every decision point.

#1	[FinaBridge] Trade Request Submitted — TR-2025-0847	
TO: Ahmed (Importer) TRIGGER: Trade Submitted - Reference: TR-2025-0847 Goods: Electronic Components Value: USD 50,000 - Instrument: Letter of Credit (LC) Submitted: 22 March 2025, 10:42 AM GST - AI verification is in progress. You will be notified of the outcome shortly.		
#2	[FinaBridge Internal] New Trade Submitted — TR-2025-0847 AI Processing	
TO: Macy + Farah + Reda (simultaneously) TRIGGER: Trade Submitted - Importer: Ahmed Al-Rashid Trading LLC Value: USD 50,000 Instrument: LC - AI engine is running. No action required at this stage. - You will receive the AI verification result within 2 minutes.		
#3A	[FinaBridge] AI Passed — TR-2025-0847 ACTION REQUIRED: Macy	
TO: Macy + Farah + Reda TRIGGER: AI Verification PASSED - Fraud Score: 8/100 — LOW RISK All fields verified No flags raised - Extracted: Bank of China LC2025-88442 USD 50,000 Exp: 30 Sep 2025 KYC Match - Macy: please complete Tier 1 review within 1 business day. [Open Admin Panel]		
#3B	[FinaBridge] AI Verification FAILED — TR-2025-0847 Rejected	
TO: Macy + Farah + Reda + Ahmed TRIGGER: AI Verification FAILED - Score: 78/100 HIGH RISK Flags: Amount mismatch (\$30K vs \$50K) + Document expired (15 Jan 2025) - Request automatically rejected. Importer has been notified separately. - No admin review action required. Logged for audit purposes.		
#4	[FinaBridge] LC Verified — Your Request is Under Admin Review	
TO: Ahmed (Importer) TRIGGER: AI Passed - Your Letter of Credit has successfully passed automated compliance verification. - Your trade request is now progressing through our internal admin review. - Estimated review time: 1–2 business days. You will be notified at each stage.		
#5	[FinaBridge] Tier 1 Approved — TR-2025-0847 ACTION REQUIRED: Farah	
TO: Farah (ACTION) CC: Reda, Macy TRIGGER: Macy Approves (Tier 1) - Reviewed by: Macy (Operations Analyst) Timestamp: 22 March 2025, 02:15 PM GST - Note: "Document verified. Bank stamp clear. Amount and beneficiary confirmed." - Farah: please complete Tier 2 compliance review within 1 business day. [Open Panel]		
#6	[FinaBridge] Tier 2 Approved — TR-2025-0847 FINAL DECISION REQUIRED: Reda	
TO: Reda (ACTION) CC: Farah, Macy TRIGGER: Farah Approves (Tier 2) - Full Trail: AI (8/100) Macy (22 Mar 02:15) Farah (22 Mar 04:30) - Farah note: "AML clear. No sanctions match. Company registration confirmed." - Reda: your approval will publish this trade live to all exporters. [Open Final Panel]		

10 Email Notification Map (continued)

Emails 7–13: Trade approval, proposals, deal room and settlement notifications

#7	[FinaBridge] Ø<B% Trade Request APPROVED — TR-2025-0847 is Live
TO:	Ahmed (Importer)
	- Your trade has been reviewed and approved by our compliance team. - TR-2025-0847 Electronic Components USD 50,000 Status: ' OPEN — Live on Marketplace - Registered exporters can now view and submit proposals. [View Your Trade]
#7B	[FinaBridge Internal] Director Approved — TR-2025-0847 Published to Marketplace
TO:	Macy + Farah (FYI)
	- Director Reda has approved trade TR-2025-0847. Trade is now live. - This is an awareness notification — no action required.
#8	[FinaBridge] New Proposal Received — TR-2025-0847
TO:	Ahmed (Importer)
	- Wang Electronics Co. Ltd (Shanghai) has submitted a proposal for your trade. - Quote: USD 48,500 Timeline: 35 Days Insurance: Included FOB Shanghai - Estimated Delivery: 28 August 2025. [Review Proposal in FinaBridge]
#9	[FinaBridge] Deal Room Open — Gold Locked — TR-2025-0847
TO:	Ahmed (Importer)
	- You accepted Wang Electronics Co. Ltd's proposal for USD 48,500. 15.234g gold has been locked in Settlement Hold (cryptographic reserve). - A secure Deal Room is now open. Begin document coordination. [Enter Deal Room]
#10	[FinaBridge] Your Proposal Was Accepted — Enter Deal Room
TO:	Wang (Exporter — David Wang)
	- Congratulations. Ahmed Al-Rashid has accepted your proposal for TR-2025-0847. - Gold settlement (15.234g) is locked and ready for release upon confirmed delivery. - Please enter the Deal Room to begin shipment coordination. [Enter Deal Room]
#11	[FinaBridge] Trade Completed — Settlement Processed — TR-2025-0847
TO:	Ahmed (Importer)
	- Your FinaBridge trade has been successfully completed and settled. - Gold Settlement: 15.234g transferred to exporter Platform Fee: 0.152g - Completed: 15 September 2025, 09:00 AM GST. [Download Trade Certificate]
#12	[FinaBridge] Gold Settlement Received — TR-2025-0847
TO:	Wang (Exporter — David Wang)
	15.082g gold (net of platform fee) has been credited to your FinaBridge Wallet. - Trade reference TR-2025-0847 is now marked Complete. - You may convert or withdraw your gold via FinaPay. [Go to FinaPay]
#13	[FinaBridge] Deal Room Closed — TR-2025-0847 Archive Available
TO:	Ahmed + David Wang (Both Parties)
	- The Deal Room for trade TR-2025-0847 has been formally closed by Finatrades admin. - Closed by: Finatrades Admin Team Closed at: 15 September 2025, 10:30 AM GST - Closure Note: "Trade settled. All documents verified. Gold transferred successfully." - All communications and documents are archived and remain accessible. [View Archive]

#	Trigger	Recipient(s)	Type
1	Submitted	Ahmed	User '
2	Submitted	Macy+Farah+Reda	Internal &'
3A	AI Passed	Macy+Farah+Reda	Action &

11 Technology Architecture

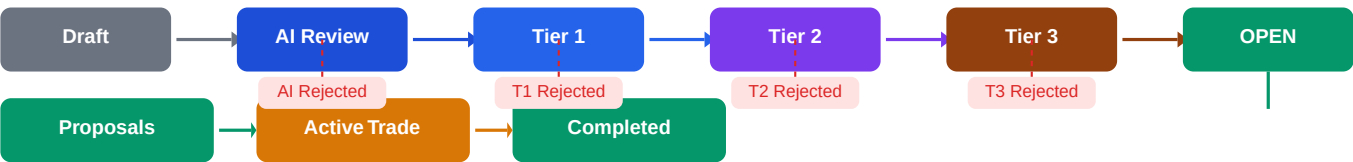
AI stack, infrastructure, storage and real-time communication components

AI & Document Processing Stack

AI Vision Engine	OpenAI GPT-4o Vision API	Multi-modal document reading — processes text, tables, stamps, seals, handwriting and signatures simultaneously from LC/POL/WR scans
OCR Secondary Layer	OCR.space Free API (500 req/day)	Fallback text extraction for documents where vision model needs supplementary OCR confirmation — zero cost tier
Job Queue	BullMQ + Redis (Upstash)	Async AI processing — document analysis runs in background without blocking user session. Auto-retry on failure. Priority queue for urgent reviews.
File Storage	Cloudflare R2	Encrypted document storage — S3-compatible API. Direct URL access for AI analysis. Current bucket: finatrades-gold. All uploads persist permanently.
Database	PostgreSQL on AWS RDS (ap-southeast-1)	Singapore region. Trade records, AI results, settlement holds, audit logs, deal room messages — all in one schema. SSL enforced.
Real-time Chat	Socket.IO (WebSocket)	Deal Room messaging with live presence indicators, typing status and instant message delivery. Rooms scoped by deal room ID.
Email Service	Nodemailer (existing)	13 email templates triggered by status changes. HTML templates with Finatrades branding. Admin CC logic for internal awareness emails.
Platform	Node.js / Express + React 18 + Vite	TypeScript throughout. Drizzle ORM for database. Tailwind v4 + shadcn/ui components. Authentication via session middleware.

12 Status Flow Diagram

Complete trade lifecycle — all states and transition paths



13 Security, Compliance & Implementation Roadmap

Regulatory framework and phased delivery plan with effort estimates

Compliance & Legal Framework

Anti-Money Laundering (AML)	Tier 2 Farah performs dedicated AML screening on every trade before it reaches the Director AI engine flags any structuring indicators (e.g., split amounts, unusual patterns) as high-risk
	All AML decisions logged with reviewer identity, timestamp and reasoning for regulatory audit
Know Your Customer (KYC)	Importer KYC name automatically cross-matched against LC/POL/WR beneficiary field by AI Any name mismatch flagged as high-risk fraud indicator requiring manual override to proceed
	Full verified KYC required before FinaBridge access is granted to any user
Sanctions & Geo-Restrictions	Farah (Tier 2) performs sanctions list and PEP screening on both trading parties Country-level geo-restrictions applied at platform level — blocked jurisdictions auto-reject
	No manual bypass possible without Director-level override with written justification
Dispute Resolution	All Deal Room Terms explicitly reference Dubai International Arbitration Centre (DIAC) Complete message history with timestamps preserved as legally admissible evidence
	Document metadata (upload time, IP, user) retained for 7 years per UAE commercial law

Implementation Roadmap — Four Phases

Schema & Form Changes		
	- New field: paymentInstrumentType (LC/POL/WR/FinaBridge Wallet) - New fields: supportingDocumentUrl, aiVerificationStatus, aiExtractedData, tier1/2/3 status - Status enum: add AI Review, AI Rejected, Tier 1/2/3 Review values - FinaBridge form UI: replace Finance Type with Payment Instrument selector - Conditional document upload zone (appears only for LC/POL/WR selection)	
AI OCR Service		
	- Build server/services/ocr-service.ts — central AI document analysis engine - BullMQ job queue for asynchronous non-blocking document processing - GPT-4o Vision integration with document-type-specific prompts (LC/POL/WR) - Fraud scoring engine: 6-point check (amount, expiry, KYC, bank, consistency, format)	

Tier 1/2/3 queue tabs in FinaBridgeManagement.tsx admin page

- Approve/Reject actions with mandatory review notes for each tier
- Role-based access: Macy sees Tier 1 only, Farah sees Tier 2 only, Reda sees all
- Full review trail display on each request card (AI + Macy + Farah notes)
- publishedToExporters flag trigger on Reda approval !' status = OPEN

13 professional HTML email templates with full Finatrades branding

- Trigger hooks at every status change point in routes.ts
- Admin team emails: Macy, Farah, Reda — simultaneous on submission, AI result, each tier
- CC logic: Macy CC'd on Tier 2 approval; Farah+Macy CC'd on Tier 3 approval
- Importer/Exporter emails: submission, AI result, approval/rejection, deal room, settlement

Ready for Chairman Approval

This document provides the complete technical, operational and compliance blueprint for the FinaBridge Option D Facilitator Model.

- ' Full AI document verification pipeline with 6-point fraud scoring
- ' Structured 3-tier human review: Macy (Operations) ! Farah (Compliance) ! Reda (Director)
- ' 13 automated email notifications covering every touchpoint in the complete journey
- ' Gold-locked Settlement Hold — neutral custody with cryptographic reserve
- ' Secure real-time Deal Room with permanent audit trail
- ' Full regulatory compliance: AML, KYC, Sanctions, DIAC arbitration, UAE law
- ' Implementation ready in 4 phases — estimated 16–23 development days total

Prepared by Finatrades Technology & Compliance Division

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